Computer Vision
Exam A

September 2025

Name, Surname, Student ID [please compile here]:

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The exam has to be carried out in 1 hour.

Exercise 1. [8 points]
Reasoning question: Consider two alternative approaches for segmenting a video into visually similar segments:

- Approach A: Use pixel intensity histograms as features and Euclidean distance as a similarity measure, followed by k-means clustering.
- Approach B: Use deep CNN-based frame embeddings as features and cosine similarity as a distance measure, followed by a clustering method.

Compare the two approaches in terms of robustness, computational efficiency and scalability. Discuss in which scenarios Approach A might outperform Approach B, and vice versa. How the choice of features could affect the quality of segmentation?

Exercise 2. [8 points]
Explain the concept of padding in the context of convolutional image filtering. How does the choice of padding method influence edge detection results, especially near the image boundaries?

Exercise 3. [8 points]
Describe a method for estimating optical flow in image sequences. What is the aperture problem, and how does it affect motion estimation? In which direction within the image is optical flow ambiguous due to this problem?

Exercise 4. [8 points]
a) What is the relationship between the fundamental matrix F and the essential matrix E? Explain how one can be derived from the other.
b) Describe how the fundamental matrix F can be estimated from corresponding points in two images. What method(s) are commonly used for this computation?